COLLABORATION NETWORKS IN THE MUSIC INDUSTRY: A NESTED ERGM ANALYSIS

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Research Question

Music collaborations form complex social networks shaped by artistic similarity, industry dynamics, and shared professional connections. Understanding these collaboration patterns provides insight into how creative partnerships emerge within the music industry.

What factors shape collaboration patterns among popular music artists?

- Do artists from the same country collaborate more frequently?
- Does genre similarity influence collaborations?
- Does network structure increase the likelihood of collaboration formation?

Dataset

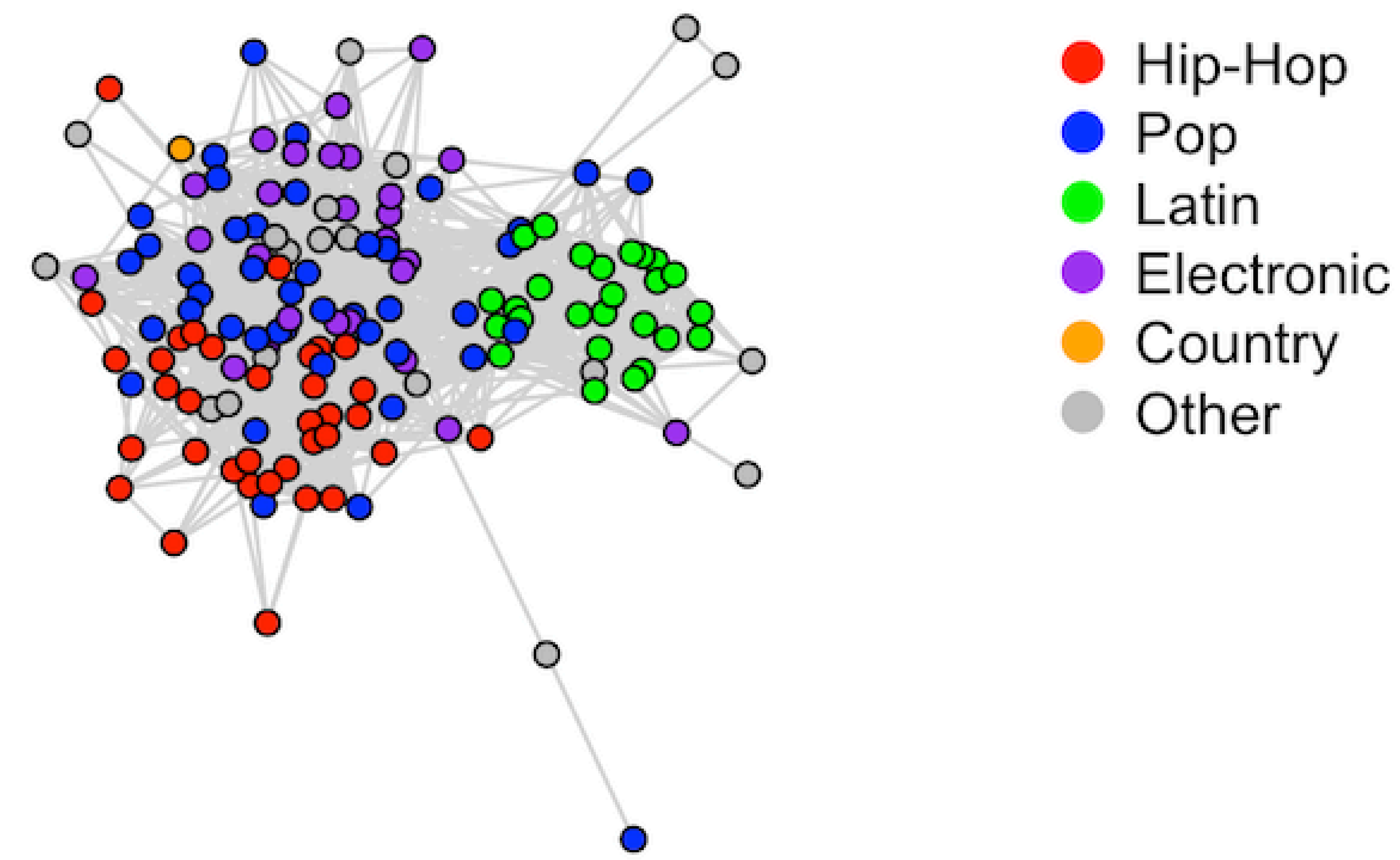
- Publicly available Kaggle music collaboration dataset
- Top 150 most collaborative artists selected
- Undirected artist collaboration network constructed
- Nodes represent artists and edges represent collaborations
- Artist attributes included:
  - Country
  - Genre group
  - Collaboration count

Hypotheses

- Artists from the same country are more likely to collaborate with each other.
- Artists sharing similar musical genres are more likely to collaborate.
- Artists who share collaborators are more likely to form collaborations themselves.
- Highly collaborative artists are more likely to form additional collaborations within the network.

Network Summary

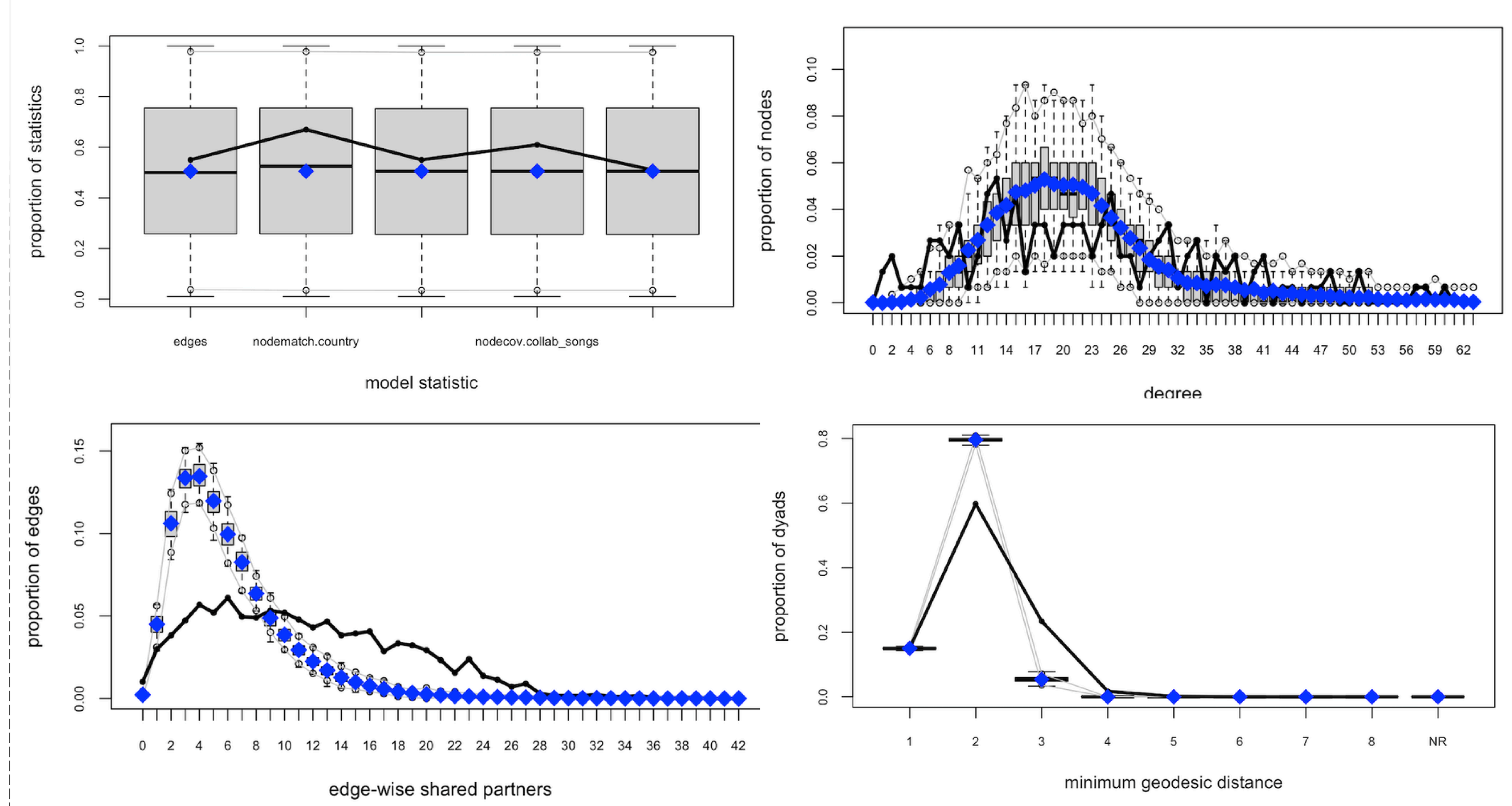
- The final collaboration network contained 150 artists connected through collaboration ties.
- The network formed a single large connected component, indicating a highly interconnected collaboration structure.
- Visible clustering patterns suggest collaborations are influenced by genre similarity and shared connections.



Methodology

An Exponential Random Graph Model (ERGM) was used to examine collaboration tie formation within the artist network. The model incorporated structural and attribute-based effects, including country homophily, genre similarity, collaboration activity, and triadic closure through GWESP terms. Nested ERGM specifications were estimated incrementally to evaluate improvements in model fit and network representation.

Goodness-of-Fit



GOF diagnostics demonstrated that the final ERGM adequately reproduced key structural properties of the observed collaboration network, including degree distributions, shared partner structures, and geodesic distances.

ERGM Results

Term	Estimate	Interpretation
Country Homophily	+0.895	Same-country artists collaborated more frequently
Genre Homophily	+0.995	Artists sharing genres were more likely to collaborate
Collaboration Count	+0.003	Highly collaborative artists formed additional ties
GWESP	+1.862	Strong clustering and triadic closure were present

Key Findings

- Artists from the same country were significantly more likely to collaborate.
- Genre similarity strongly influenced collaboration formation.
- Highly collaborative artists formed additional connections within the network.
- Collaboration networks exhibited strong clustering and triadic closure behavior.

Conclusion

The nested ERGM successfully captured both structural and attribute-based mechanisms underlying artist collaborations. Results suggest that collaboration patterns in the music industry are influenced by shared characteristics such as country and genre, as well as endogenous network processes including clustering and repeated partnership formation.

Future Work

- Incorporating temporal dynamics to examine how collaboration patterns evolve over time
- Expanding the network to include additional artist attributes such as record labels and streaming popularity
- Modeling weighted collaborations based on collaboration frequency or commercial success of tracks